SEBASTIAN H. M. HICKMAN

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EDUCATION

University of Cambridge

PhD, AI for Environmental Risks (AI4ER CDT) 2021 - 25
Understanding ozone, climate and their interactions with causal machine learning
Fully funded by the Engineering and Physical Sciences Research Council.

MRes, AI4ER CDT 2020 - 21
Probabilistic Machine Learning, Machine Learning in the Physical World, Advanced Machine Learning
Formal graduate-level courses and training in modern ML.

MSci, Natural Sciences (Physical Chemistry) 2019 - 20
Computer Simulation Methods in Physics, Atmospheric Chemistry and Global Change
Earth System and Climate Change, Energy Landscapes and Soft Matter, Polymers, Diffraction

BA, Natural Sciences 2016 - 19
Maths, Physics, Chemistry, Earth Sciences, Scientific Computing

RESEARCH EXPERIENCE

WeatherGenerator: observation-driven foundation modelling for weather and climate Jun. 2025 -
Machine Learning Scientist at the European Centre for Medium-Range Weather Forecasts, UK

- Developing state-of-the-art probabilistic generative models for Earth system modelling and forecasting
- Processing and training with terabytes of observational weather data
- Implementing efficient distributed training of multi-billion parameter models

Causal ML for seasonal forecasting and climate model emulation Mar. to Sep. 2024
Intern with Prof. David Rolnick, Mila – Quebec AI Institute, Canada

- Developed probabilistic generative models for seasonal weather forecasting and climate model emulation
- Algorithmic improvement of dynamic variational autoencoder architectures in PyTorch

AtmoRep evaluation and development Sep. to Nov. 2023
Helmholtz Visiting Researcher with Prof. Martin Schultz, Forschungszentrum Jülich, Germany

- Testing and evaluating the AtmoRep foundation model and codebase
- Large-scale computing on some of Europe's most powerful HPC facilities

Research in machine learning for climate science Oct. 2021 to Apr. 2025
PhD student at the University of Cambridge, UK

- Researching, building, and deploying ML for forecasting ozone and climate, published in top-tier venues
- Implemented efficient distributed training of transformers and probabilistic generative models in PyTorch
- Wrote initial code and contributed to a widely used open-source Python package, detectree2

TECHNICAL SKILLS

Languages Python, Julia, R, html, French, basic Fortran
Libraries PyTorch, xarray, zarr, netcdf, dask, PyTorch Lightning, scikit-learn, NumPy, pandas, DoWhy

SELECTED PUBLICATIONS

Hickman, S.H.M., J. Boussard, C. Lange, I. Trajkovic, Griffiths, P. T., Nowack, P. J., Archibald, A.T., Gurwicz, Y., Rolnick, D., 2025. Causal Climate Emulation with Bayesian Filtering.
NeurIPS.

Kelp, M. M., **Hickman, S.H.M.**, Griffiths, P. T., Miyazaki, K., Pennington, E. A., Koren, G., Iglesias-Suarez, F., Schultz, M. G., Chang, K., Cooper, O. R., Zhu, Q., 2025. AI4O3: A Foundational Data Collection for Artificial Intelligence in Tropospheric Ozone Research
NeurIPS AI4Science Workshop.
Renaissance Philanthropy Prize.

Hickman, S.H.M.*, Kelp*, M. M., Griffiths*, P. T., Doerksen*, K., Miyazaki*, K., Pennington*, E. A., Koren*, G., Iglesias-Suarez*, F., Schultz*, M. G., Chang, K., Cooper, O. R., Archibald, A. T., Sommariva, R., Carlson, D., Wang, H., West, J. J. and Liu, Z, 2025. Applications of Machine Learning and Artificial Intelligence in Tropospheric Ozone Research.
Geoscientific Model Development.
Highlight paper.

Hickman, S.H.M., Griffiths, P. T., Nowack, P. J. and Archibald, A.T., 2023. Short-term forecasting of ozone air pollution in Europe with transformers.
Environmental Data Science.
Also in NeurIPS Tackling Climate Change with Machine Learning Workshop, 2022.

Hickman, S.H.M.*, Ball, J.G.C.*, Jackson, T.D.*, Koay, X.J., Hirst, J., Jay, W., Aubry-Kientz, M., Vincent, G. and Coomes, D.A., 2023. Accurate delineation of individual tree crowns in tropical forests from aerial RGB imagery using Mask R-CNN.
Remote Sensing in Ecology and Conservation.

SELECTED AWARDS AND GRANTS

NeurIPS AI4Science Dataset Proposal Prize 2025
Prize awarded by Renaissance Philanthropy for the best scientific dataset proposals.

Helmholtz Visiting Researcher Grant 2023
Grant from the Helmholtz Information and Data Science Academy to visit PD Prof. Martin Schultz to work on AtmoRep, a foundation model of atmospheric dynamics.

Enrichment Scheme 2022-23
Awarded a place on the Alan Turing Institute's Enrichment Scheme.

Frisby Fund & Hawks' Trust Grants 2017-20
Corpus Christi College & The Hawks' Trust
Awarded for academic and sporting achievement as a Cambridge undergraduate.

ACTIVITIES

Co-Chair, Machine Learning for Tropospheric Ozone (ML4O3) Group 2023 - 2025

- Coordinating a team of academics to write a perspective paper on machine learning for modelling ozone
- Developed excellent interpersonal relationships and strong communication with leading academics

President, Cambridge University Artificial Intelligence Society 2021 - 22

- Led CUAI, a university society of 1200 members. Secured society sponsorship for the first time, raising £25k total from organisations including DeepMind, Microsoft Research and Amazon Research.

Captain, Cambridge University Golf Club 2018 - 19

- Directed CUGC, one of the nine Full Blue clubs at Cambridge. Organised the activities of the club including funding, fixtures and training, managing £30k of funds and securing an increase in total funding.